

Project Genesis: Technical Architecture Reference

Phase 8.4 Implementation Reference

Vinay Khosya

Project Genesis Research Laboratory

vinaykhosya/project-genesis June 2026

Abstract—This document is the technical implementation reference for Project Genesis at Phase 8.4. It describes the repository layout, the `WorldState` unified state container, the world generation pipeline, the agent class architecture, the drive and emotion system, the cognitive planning stack, memory subsystems, genetics and reproduction, telemetry, and configuration schema. It is intended for developers and contributors who need to understand the current codebase. For research hypotheses, experimental results, and scientific discussion, refer to *genesis_monograph.tex*. For the chronological development diary, refer to *genesis_lab_notebook.tex*.

Index Terms—artificial life, agent architecture, world generation, evolutionary simulation, Python, NumPy, terrain generation, cognitive architecture.

I. REPOSITORY LAYOUT

The project root is structured into five main areas, shown in Figure 1.

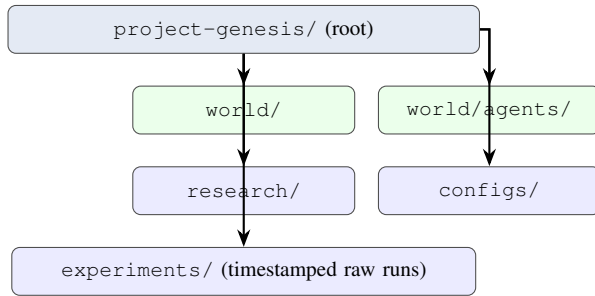


Fig. 1. Top-level project structure (Phase 8.4).

Key entry-point scripts are summarised in Table I.

TABLE I
ENTRY-POINT SCRIPTS

File	Purpose
main.py	World generation + interactive sim loop
run.py	Headless run with full telemetry export
run_research.py	Ablation sweep experiment runner
run_resume.py	Resume from JSON snapshot
run_test.py	Unit and integration test runner
visualizer.html	Web-based real-time dashboard

II. WORLDSTATE — UNIFIED STATE CONTAINER

All simulation state is centralised in one `WorldState` instance (`world/state.py`) passed through every pipeline

step and mutated in-place. No global variables are used. Table II lists the primary grid arrays; all are 2D NumPy arrays of shape `(height, width)`.

TABLE II
WORLDSTATE GRID ARRAYS

Field	dtype	Description
elevation	float32	Terrain height $\in [0, 1]$; sea level = 0.30
temperature	float32	Degrees Celsius (latitude + lapse rate)
rainfall	float32	mm/year (wind moisture advection)
biome	int32	Whittaker code 0–8
river_map	float32	Priority-Flood flow accumulation
lake_map	float32	Lake surface mask $[0, 1]$
wood	float32	Wood resource density $[0, 1]$
stone	float32	Stone resource density $[0, 1]$
copper	float32	Copper ore density $[0, 1]$
iron	float32	Iron ore density $[0, 1]$
wildlife	float32	Huntatable food density $[0, 1]$
fertility	float32	Soil fertility (drives replenishment)
water_score	float32	Habitability: water proximity
food_score	float32	Habitability: food proximity
resource_score	float32	Habitability: mineral resources
climate_score	float32	Habitability: thermal comfort
terrain_score	float32	Habitability: slope passability
habitability	float32	Weighted composite score $[0, 1]$
movement_cost	float32	Traversal cost (slope + biome)
death_density	float32	Accumulated deaths per cell

A. Scalar and List State

In addition to grid arrays, `WorldState` holds scalar and list fields listed in Table III.

B. `query_agents()` — Spatial Hash Grid

The method `query_agents(center, radius, alive_only)` returns all agents within `radius` cells of `center`. It uses a cached **spatial hash grid** (bin size = 32), reducing naïve $O(N^2)$ per-tick distance checks to $O(1)$ bin lookup followed by a small local linear scan.

Phase 8.4 performance ($N = 64$ agents, 128×128 grid):
~320 000 calls / 1000 ticks; avg. 3.2 agents returned;
< 0.01 ms/call.

III. WORLD GENERATION PIPELINE

The world generation pipeline runs once at startup and is entirely deterministic given a fixed seed. Figure 2 shows the ten sequential stages.

TABLE III
WORLDSTATE SCALAR & LIST FIELDS

Field	Type	Description
tick	int	Current simulation tick
agents	list	All Agent objects (alive + dead)
colonies	list	Colony metadata dicts
max_population	int	Carrying capacity cap (default 64)
generation_number	int	Highest generation seen
total_births	int	Cumulative birth count
total_deaths	int	Cumulative death count
population_history	list	Sampled per-tick census records
genetic_history	list	Sampled gene means and variances
extinction_events	list	{tick, colony_id, name}
events_timeline	list	Emergence flags and anomalies
telemetry	obj	TelemetryCollector instance

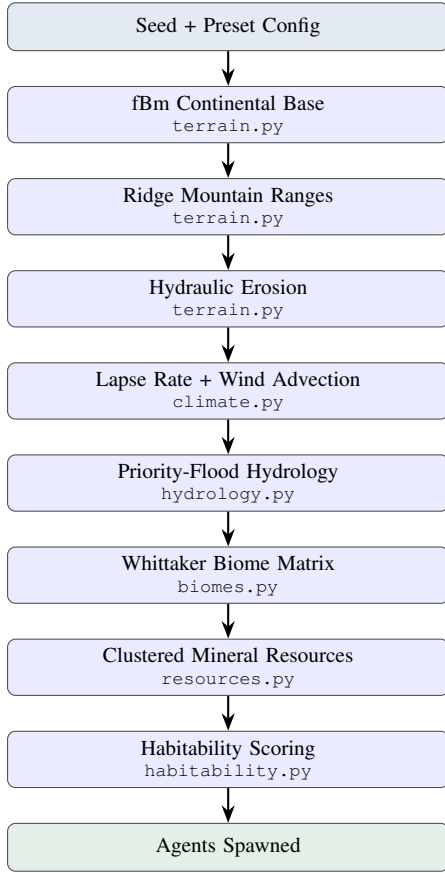


Fig. 2. World generation pipeline (Phase 8.4). Each step reads and mutates WorldState in-place.

A. Resolution-Independent Climate Scaling

Moisture advection rates scale dynamically with grid width w to keep geographic structure invariant to resolution:

$$P_{\text{base}} = \frac{0.6}{w}, \quad e_{\text{ocean}} = \frac{50.0}{w}, \quad e_{\text{land}} = \frac{8.0}{w} \quad (1)$$

B. Sea Level Wind Clamping

Before computing advection slope, elevation is clamped to sea level to ensure prevailing winds blow across flat water:

$$H_c(y, x) = \min(H(y, x), 0.30) \quad (2)$$

IV. AGENT ARCHITECTURE

A. Agent Class Fields

The Agent class (`world/agents/agent.py`) groups its fields into six logical sections, summarised in Table IV.

TABLE IV
AGENT CLASS FIELD GROUPS

Section	Key Fields
Identity	id, location, home_location
Genome	genome (14-gene), brain (expressed phenotype)
Physiology	health, hunger, thirst, energy, injury, thermal_stress
Age	age, max_age, life_stage
Inventory	stored_food, stored_water, canteen, shelter_level (0–2)
Drives	drives (biological tensions), emotions (fear/frustration/...), motivation (6-dim. profile)
Cognition	memory (episodic), knowledge (spatial), predictor (MLP), action_queue
Social	colony_id, relationships, mates, children
Lifecycle	dead, cause_of_death, generation, parent_ids

B. Life Stages

TABLE V
AGENT LIFE STAGES

Stage	Age (years)	Notes
Infant	0–3	Cannot reproduce; lower metabolic rate
Juvenile	3–14	Cannot reproduce; full exploration
Adult	14–60	Full reproduction eligibility
Elder	60+	Senescence factor active

V. DRIVE & EMOTION SYSTEM

A. Clock-Tiered Updates

Four processing tiers operate at different timescales (Table VI).

B. Welford’s Online Motivation Drift

Every 100 ticks, motivational priority weights drift toward the agent’s chronic drive history using Welford’s algorithm [1]:

TABLE VI
DRIVE SYSTEM UPDATE SCHEDULE

Layer	Interval	Contents
Biological Tensions	every tick	Hunger, thirst, energy, injury, thermal
Emotional Dynamics	$\div 10$	Fear, frustration, longing, grief, boredom
Relationship Graph	$\div 30$	Trust, respect, attachment, competition
Motivation Profile	$\div 100$	6-dim personality weights (Welford drift)

$$n \leftarrow n + 1 \quad (3)$$

$$\delta \leftarrow x - \mu \quad (4)$$

$$\mu \leftarrow \mu + \delta/n \quad (5)$$

$$M_2 \leftarrow M_2 + \delta(x - \mu) \quad (6)$$

$$\sigma^2 = M_2/(n - 1) \quad (7)$$

No stored history buffer is required; the algorithm runs in $O(1)$ time and $O(1)$ memory per agent per update.

C. Lateral Inhibition Conflict Resolver

When multiple motivational drives are simultaneously elevated, mutual suppression prevents behavioral oscillation:

$$M_i = \max \left(0, m_i - \beta \sum_{j \neq i} m_j \right), \quad \beta = 0.15 \quad (8)$$

D. Relationship Trust

Every 30 ticks, episodic memory is reviewed and dyadic trust $T_{ij} \in [-1, 1]$ updated:

$$\delta = 0.10 n_s + 0.15 n_m - 0.20 n_d \quad (9)$$

$$T'_{ij} = \begin{cases} T_{ij} + \delta(1 - T_{ij}) & \delta \geq 0 \\ T_{ij} + \delta(1 + T_{ij}) & \delta < 0 \end{cases} \quad (10)$$

VI. COGNITIVE PLANNING STACK

A. Per-Tick Processing Order

Figure 3 illustrates the ten-stage per-agent tick cycle.

B. Utility Computation

For each candidate action a , the final utility is:

$$U_a = \underbrace{\sigma(N; \theta_a, s_a)}_{\text{sigmoid base}} \times \underbrace{\mathbf{w}_e \cdot \mathbf{f}_{\text{env}}}_{\text{env mod}} \times \underbrace{\sum_k M_k \alpha_{ak}}_{\text{drive mod}} + \underbrace{\hat{r}(a)}_{\text{predictor}} \quad (11)$$

where σ is the sigmoid base utility (12), M_k are the laterally inhibited motivational weights, and $\hat{r}$ is the neural predictor reward estimate.

$$\sigma(N; \theta, s) = \frac{100}{1 + e^{-(N-\theta)/s}} \quad (12)$$

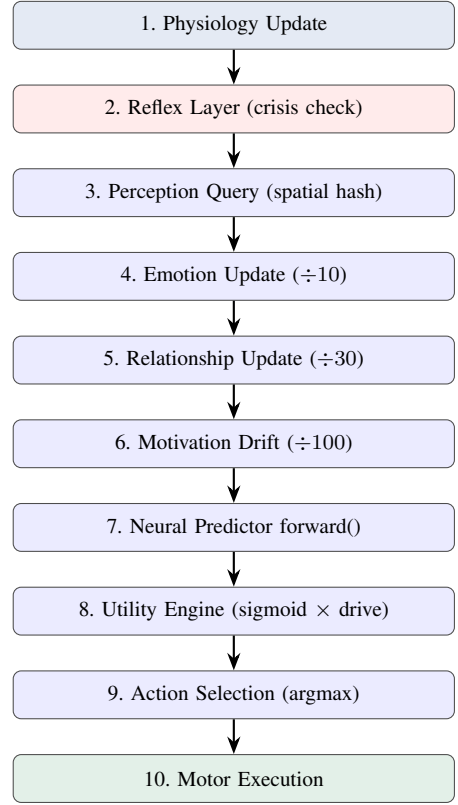


Fig. 3. Per-agent tick processing order (Phase 8.4). The Reflex Layer (step 2) bypasses the cognitive stack for metabolic emergencies.

C. Online Neural Predictor ($20 \rightarrow 3 \rightarrow 1$)

$$\mathbf{h} = \tanh(W_1 \mathbf{x} + \mathbf{b}_1), \quad \hat{r} = W_2 \mathbf{h} + b_2 \quad (13)$$

Online backpropagation (after each outcome):

$$\varepsilon = r - \hat{r}, \quad \Delta W_2 = \eta \varepsilon \mathbf{h}, \quad \Delta W_1 = \eta \varepsilon (W_2^T \circ (1 - \mathbf{h}^2)) \mathbf{x}^T \quad (14)$$

Gradient explosion is prevented by a sleep-based weight consolidation buffer (Phase 7.5): weights update from a batch of the 20 most recent outcomes only when the agent enters rest state.

VII. GENETICS & REPRODUCTION

A. 14-Gene Genome

Table VII lists all genes and their expressed phenotypic effects.

B. Genetic Operators

Mendelian crossover and Gaussian mutation:

$$g_i^c = \begin{cases} g_i^A & p = 0.5 \\ g_i^B & p = 0.5 \end{cases}, \quad g_i^c \leftarrow \text{clip}(g_i^c + \mathcal{N}(0, 0.05^2), 0, 1) \quad (15)$$

with per-gene mutation probability $P_{\text{mut}} = 0.05$.

TABLE VII
14-GENE GENOME (PHASE 8.4)

Idx	Gene	Phenotypic Effect
0	$g_{\text{metabolism}}$	Metabolic need decay rate
1	$g_{\text{thermoreg}}$	Thermal stress resistance
2	g_{vision}	Vision radius (4–16 cells)
3	g_{mobility}	Movement energy efficiency
4	$g_{\text{mem_fidelity}}$	Spatial memory decay rate
5	g_{planning}	Near vs. far resource bias
6	g_{novelty}	Exploration drive weight
7	g_{social}	Kin proximity utility
8	$g_{\text{aggression}}$	Territory size & dispute rate
9	g_{sharing}	Food/water donation utility
10	g_{risk}	Combat retreat threshold
11	$g_{\text{resilience}}$	Injury resistance
12	$g_{\text{longevity}}$	Maximum lifespan modifier
13	g_{learning}	Neural predictor learning rate

C. Reproduction Eligibility Gates

An agent may reproduce only when all five conditions hold simultaneously:

- 1) Adult life stage (≥ 14 years);
- 2) Colony size $< N_{\text{max}} = 64$;
- 3) Hunger < 70 and Thirst < 70 ;
- 4) $T_{ij} > 0.2$ with candidate mate;
- 5) Reproduction cooldown timer = 0.

VIII. TELEMETRY & OBSERVABILITY

Table VIII lists the data streams collected by TelemetryCollector (world/telemetry.py).

TABLE VIII
TELEMETRY DATA STREAMS

Stream	Rate	Content
Population census	$\div 10$	Total + per-colony alive count
Genetic history	$\div 50$	14-gene means, variances, diversity
Death traces	per event	Cause, location, age, agent ID
Dispute logs	per event	Aggressor, defender, outcome
Path history	per step	Agent movement coordinates
Dehydration traces	per death	Full agent state at dehydration
Emergence events	per event	Colony splits, settlements, crashes
Causal chains	per hazard	Trigger→state→resolution

A. Export Formats

- `live_state.json / .js` — web dashboard feed
- `agent_census.csv` — spreadsheet analysis
- `evolution_journal.md` — human-readable genetic drift log
- `research/EXP-*/report.md` — per-experiment scientific reports

IX. CONFIGURATION SCHEMA

All experiments are configured via JSON (`configs/default_experiment.json`). Table IX describes the ablation flags.

TABLE IX
ABLATION FLAG REFERENCE

Flag	true (ON)	false (ABLATED)
<code>planner</code>	Full utility evaluation	Random action selection
<code>emotion</code>	EMA emotional drives	Drives clamped to 0.0
<code>relationships</code>	Trust/respect graph	All social weights neutral
<code>memory_imp.</code>	Genome-gated importance	Uniform exponential decay
<code>motivation</code>	Lateral inhibition + drift	Fixed uniform weights
<code>pred_error</code>	Online MLP learning	Prediction error zeroed

X. PERFORMANCE CHARACTERISTICS

Measured on a standard laptop (Intel i7, 16GB RAM, Python 3.11) at default settings (128×128, 64 agents):

TABLE X
PHASE 8.4 PERFORMANCE MEASUREMENTS

Metric	Value
Ticks per second	12–18
WorldState memory	≈ 45 MB
World generation time	3–6 seconds
Telemetry write overhead	$\approx 8\%$ of tick time
Spatial hash rebuild	≈ 0.3 ms/tick
Peak memory (full tel.)	≈ 180 MB

A. Known Bottlenecks

- 1) **Telemetry encoding.** Default Windows Cp1252 encoding caused 15× slowdown on Unicode event logs. Fixed by forcing UTF-8 and async telemetry write queue.
- 2) **Perception fallback.** Mid-tick spatial hash invalidation falls back to $O(N^2)$ brute-force. Guards added to prevent mid-tick invalidation.
- 3) **Episodic memory growth.** Without a FIFO cap, memory lists reach 5000+ entries per agent. Resolved by rolling cap (max 200 entries).

XI. WORLD OUTPUT FIGURES

Figures 4–8 show representative outputs of the world generation pipeline for seed 1010 on a 128×128 grid.

REFERENCES

- [1] B. P. Welford, “Note on a method for calculating corrected sums of squares and products,” *Technometrics*, vol. 4, no. 3, pp. 419–420, 1962.
- [2] R. Barnes, “RichDEM: High-performance terrain analysis,” *Geomorphology*, 2016.
- [3] W. Wang and D. Liu, “An efficient method for identifying and filling surface depressions in digital elevation models,” *Int. J. Geogr. Inf. Sci.*, vol. 20, no. 2, pp. 193–213, 2006.

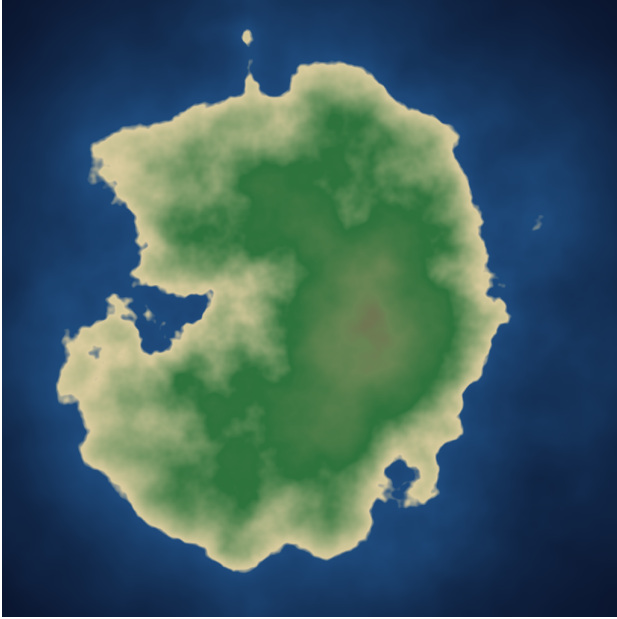


Fig. 4. Elevation map (seed 1010). Dark = deep ocean; bright ridges = mountains. Sea level = 0.30 (normalised).

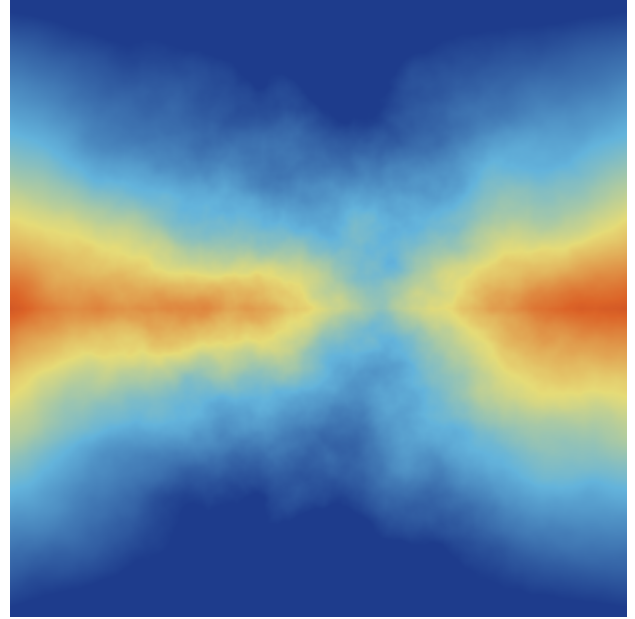


Fig. 6. Temperature ($^{\circ}\text{C}$). Equatorial band is warmest; poles are coldest. High-elevation cells are visibly cooler than surroundings.

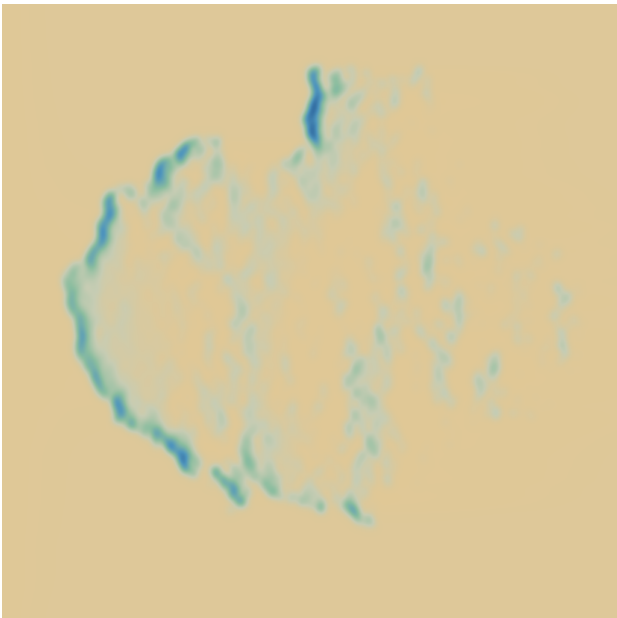


Fig. 5. Annual rainfall (mm/year). High on west-facing windward slopes; pronounced rain shadow on eastern leeward faces.



Fig. 7. Whittaker biome classification (seed 1010). Eight biome classes derived from the joint (T, R) grid.

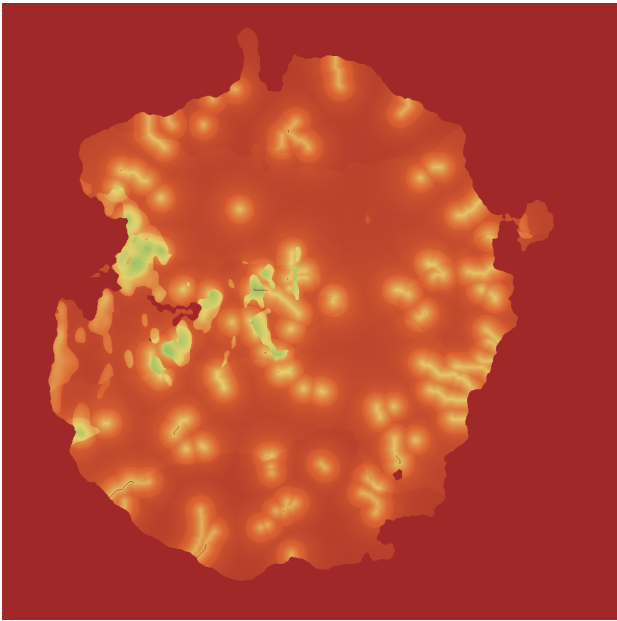


Fig. 8. Composite habitability score. Bright zones combine water proximity, food density, climate comfort, and terrain passability.